

# Linear Algebra 1A Cheat Sheet

*Convention: a matrix of size  $r \times c$  has  $r$  rows and  $c$  columns.*

## 1. Fields

A field  $\mathbb{F}$  is a set with operations  $(+, \cdot)$  satisfying: closure, associativity, commutativity, identities  $(0, 1)$ , inverses, and distributivity.

**Characteristic.**

- $\text{char}(\mathbb{F}) :=$  the least  $n \geq 1$  with  $\overbrace{1 + \cdots + 1}^{n \text{ times}} = 0$ , and  $\text{char}(\mathbb{F}) := 0$  if no such  $n$  exists
- $\text{char}(\mathbb{F})$  is always 0 or prime
- $\forall \mathbb{F} : \text{char}(\mathbb{F}) = p > 0 \Rightarrow \mathbb{Z}_p$  is a subfield of  $\mathbb{F}$

**Finite fields.**

- $|\mathbb{F}| < \infty \Rightarrow |\mathbb{F}| = p^r$ , where  $p = \text{char}(\mathbb{F})$  and  $r \in \mathbb{N}$
- $\forall (p \in \mathbb{P}) . \forall (r \in \mathbb{N}) . \exists (\mathbb{F} \text{ field}) : |\mathbb{F}| = p^r$

## 2. Matrices

$$A_{m \times p} B_{p \times n} = (AB)_{m \times n}, \quad (AB)_{ij} := \sum_{k=1}^p a_{ik} b_{kj}$$

- $AB = 0$  does not imply  $A = 0$  or  $B = 0$
- If  $A^T = A$  and  $B^T = B$ , then  $AB$  is symmetric if and only if  $AB = BA$
- If  $\varphi$  is an elementary row operation and  $A \in \mathbb{F}^{m \times n}$ , then

$$\varphi(A) = \varphi(I_m) A$$

## 3. Vector Spaces & Subspaces

A subset  $W \subseteq V$  is a subspace if and only if:

- $W$  is nonempty (equivalently  $0 \in W$ )
- Closed under addition: if  $u, v \in W$  then  $u + v \in W$
- Closed under scalar multiplication: if  $\alpha \in \mathbb{F}$  and  $u \in W$  then  $\alpha u \in W$

Equivalently:  $\alpha u + v \in W$  for all  $u, v \in W$  and all  $\alpha \in \mathbb{F}$ .

- $U \cap W$  is a subspace of  $V$
- $U \cup W$  is a subspace of  $V$  if and only if  $U \subseteq W$  or  $W \subseteq U$
- $U + W := \{u + w : u \in U, w \in W\} = \text{span}(U \cup W)$  is a subspace of  $V$
- Direct sum:  $U \oplus W$  if and only if  $U \cap W = \{0\}$  (general case: see Direct sums, §6)
- $\text{span}(S)$  is the smallest subspace containing  $S$
- $S_1 \subseteq S_2 \subseteq V \Rightarrow \text{span}(S_1) \subseteq \text{span}(S_2) \subseteq V$
- $W \leq V \Rightarrow \text{span}(W) = W$
- $S \subseteq W \leq V \Rightarrow \text{span}(S) \leq W$

## 4. Rank

- $\text{rank}(A_{m \times n}) \leq \min\{m, n\}$
- $\text{rank}(A^T) = \text{rank}(A)$
- $A \sim B \Rightarrow \text{rank}(A) = \text{rank}(B)$
- $\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$
- $A, B \in \mathbb{F}^{n \times n}, AB = 0 \Rightarrow \text{rank}(A) + \text{rank}(B) \leq n$

## 5. Linear Systems

$$Ax = b, \quad n \text{ variables}, \quad r = \text{rank}(A)$$

- Number of pivots  $= r$ , free variables  $= n - r$
- System is solvable if and only if  $b \in \text{Col}(A)$
- Homogeneous system:  $Ax = 0$  has solution space dimension  $n - r$

$$\{x : Ax = b\} = x_p + \ker(A)$$

## 6. Bases & Dimension

If  $\dim V = n$ :

- Any linearly independent set has size at most  $n$
- Any spanning set has size at least  $n$
- A set  $B$  with  $|B| = n$  is a basis if and only if it is linearly independent if and only if it spans  $V$
- If  $W \leq V$ , then  $W = V$  if and only if  $\dim W = \dim V$

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$$

**Direct sums.** Let  $M_1, \dots, M_s \leq V$  with

$M = M_1 + \cdots + M_s$ . The following are equivalent:

1.  $M = M_1 \oplus \cdots \oplus M_s$
2.  $\forall a \in M$ : the representation  $a = a_1 + \cdots + a_s$ ,  $a_i \in M_i$ , is unique
3.  $\exists a \in M$ : the representation  $a = a_1 + \cdots + a_s$ ,  $a_i \in M_i$ , is unique
4.  $\forall a_i \in M_i : a_1 + \cdots + a_s = 0 \Rightarrow a_1 = \cdots = a_s = 0$
5.  $E^{(i)}$  a basis of  $M_i \Rightarrow E^{(1)}, \dots, E^{(s)}$  together form a basis of  $M$
6.  $\dim M = \dim M_1 + \cdots + \dim M_s$

Note 3  $\Rightarrow$  all: one witness with a unique decomposition suffices.

## 7. Invertible Matrices

- If  $P$  is invertible, then

$$\text{rank}(PA) = \text{rank}(AP) = \text{rank}(A)$$

- If  $\alpha \neq 0$ , then

$$(\alpha A)^{-1} = \alpha^{-1} A^{-1}$$

- If  $C$  is invertible and  $B = CA$ , then

$$B \text{ invertible} \iff A \text{ invertible}$$

## Conditions Related to Invertibility

Let  $A \in \mathbb{F}^{n \times n}$ . Each item states its relation to “ $A$  is invertible”:

1.  $\implies A^{-1}$  is unique
2.  $\implies A$  has no zero row or column
3.  $\iff \text{rank}(A) = n$
4.  $\iff \det(A) \neq 0$
5.  $\iff A^T$  is invertible, with  $(A^T)^{-1} = (A^{-1})^T$
6.  $\iff A \sim I_n$
7.  $\iff \text{RREF}(A) = I_n$
8.  $\iff Ax = 0 \Rightarrow x = 0$
9.  $\iff$  For all  $b \in \mathbb{F}^n$ , the system  $Ax = b$  has the unique solution  $x = A^{-1}b$
10.  $\iff \text{Row}(A) = \mathbb{F}^n$

11.  $\iff \text{Col}(A) = \mathbb{F}^n$
12.  $\iff$  The rows of  $A$  span  $\mathbb{F}^n$
13.  $\iff$  The columns of  $A$  span  $\mathbb{F}^n$
14.  $\iff$  The rows of  $A$  are linearly independent
15.  $\iff$  The columns of  $A$  are linearly independent
16.  $\iff$  The rows of  $A$  form a basis of  $\mathbb{F}^n$
17.  $\iff$  The columns of  $A$  form a basis of  $\mathbb{F}^n$
18.  $\iff \dim \text{Row}(A) = n$
19.  $\iff \dim \text{Col}(A) = n$
20.  $\iff A$  is a product of elementary matrices
21.  $\iff \text{adj}(A)$  is invertible
22.  $\iff A$  is invertible on one side (i.e.  $AB = I$  or  $BA = I$  for some  $B$ )
23.  $\iff A$  is elementary (then  $A^{-1}$  is also elementary)

## 8. Matrix Spaces

- $\dim \text{Row}(A) = \dim \text{Col}(A) = \text{rank}(A)$
- $\dim \ker(A) = n - \text{rank}(A)$  for  $A \in \mathbb{F}^{m \times n}$
- $\text{Row}(AB) \subseteq \text{Row}(B)$
- $\text{Col}(AB) \subseteq \text{Col}(A)$

## 9. Determinants & Adjugate

- $\det(A^T) = \det(A)$
- $\det(AB) = \det(A) \det(B)$
- $\det(A^{-1}) = (\det A)^{-1}$
- $\det(\alpha A) = \alpha^n \det(A)$  for  $A \in \mathbb{F}^{n \times n}$

**Adjugate.** For  $A \in \mathbb{F}^{n \times n}$ :  $M_{ij}(A)$  is  $A$  with row  $i$  and column  $j$  deleted; cofactor  $c_{ij} := (-1)^{i+j} \det(M_{ij}(A))$ ; cofactor matrix  $C = (c_{ij})$ .

- $\text{adj}(A) := C^T$
- $\text{adj}(A)_{ij} = c_{ji} = (-1)^{i+j} \det(M_{ji}(A))$
- $\text{adj}(A) A = A \text{adj}(A) = \det(A) I_n$
- $A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$
- $\text{rank}(\text{adj}(A)) = \begin{cases} n, & \text{rank}(A) = n, \\ 1, & \text{rank}(A) = n - 1, \\ 0, & \text{rank}(A) < n - 1 \end{cases}$

**Cramer's rule.**  $\det(A) \neq 0 \Rightarrow Ax = b$  has the unique solution

$$x_k = \frac{\det(A_k)}{\det(A)}, \quad k = 1, \dots, n$$

where  $A_k$  is  $A$  with column  $k$  replaced by  $b$ .

## 10. Linear Maps

Let  $T : V \rightarrow U$  be linear.

- Kernel:  $\ker T := \{v \in V : T(v) = 0\}$
- Image:  $\text{Im } T := \{T(v) : v \in V\}$
- $T$  is injective if and only if  $\ker T = \{0\}$
- $T$  is surjective if and only if  $\dim \text{Im } T = \dim U$

$$\dim V = \dim \ker T + \dim \text{Im } T$$

## 11. Operators & Isomorphisms

- A linear map is an isomorphism if and only if it is injective and surjective
- $V \cong U$  if and only if  $\dim V = \dim U$

- For an operator  $T : V \rightarrow V$ :

$$\ker T \subseteq \ker T^2, \quad \text{Im } T \supseteq \text{Im } T^2$$

- In finite dimensions, equality implies stabilization

## 12. Matrix Representation

Let  $B = \{v_1, \dots, v_n\}$  and  $S = \{u_1, \dots, u_m\}$ .

$$[T]_B^S := ([T(v_1)]_S, \dots, [T(v_n)]_S)$$

- $[T]_B^S[v]_B = [T(v)]_S$
- $[S \circ T]_A^C = [S]_B^C [T]_A^B$
- $T$  is invertible if and only if  $[T]_B^S$  is invertible

## 13. Dual Spaces, Functionals, and Annihilators

**Concept.** Duality studies vectors indirectly via linear measurements. Instead of examining vectors themselves, we examine how all linear functionals act on them.

**Linear functionals.** A linear functional is a linear map  $f : V \rightarrow \mathbb{F}$ . It assigns a scalar to each vector and represents a linear measurement.

**Dual space.**

$$V^* := \{f : V \rightarrow \mathbb{F} \mid f \text{ is linear}\}$$

$V^*$  is a vector space. If  $\dim V = n < \infty$ , then  $\dim V^* = n$ .

**Dual basis.** If  $B = \{v_1, \dots, v_n\}$  is a basis of  $V$ , the dual basis  $B^* = \{f_1, \dots, f_n\}$  satisfies  $f_i(v_j) = \delta_{ij}$ . Each  $f_i$  extracts the  $i$ -th coordinate.

**Second dual space.** The second dual is  $V^{**} := (V^*)^*$ . There is a canonical map

$$\Phi : V \rightarrow V^{**}, \quad \Phi(v)(f) := f(v)$$

In finite dimensions, this map is an isomorphism.

**Kernel (MERHAV HA EFES).** For  $T : V \rightarrow U$ ,

$$\ker T := \{v \in V : T(v) = 0\}$$

This is a subspace of  $V$ .

**Annihilator (MEAPES).** For  $W \leq V$ ,

$$W^\circ := \{f \in V^* : f(w) = 0 \text{ for all } w \in W\}$$

This is a subspace of  $V^*$ .

**Difference between MERHAV HA EFES and MEAPES.**

- MERHAV HA EFES: vectors in  $V$  mapped to zero by a linear map
- MEAPES: functionals in  $V^*$  that vanish on a subspace

**Dimension relation.** If  $W \leq V$  and  $\dim V = n < \infty$ , then

$$\dim W + \dim W^\circ = n$$

**Dual (transpose) map.** For  $T : V \rightarrow U$  define

$$T^* : U^* \rightarrow V^*, \quad T^*(f) := f \circ T$$

Then

$$\ker T^* = (\operatorname{Im} T)^\circ, \quad \operatorname{Im} T^* = (\ker T)^\circ$$

In matrix form,

$$[T^*] = [T]^T$$

## 14. Vandermonde Matrix

Given scalars  $x_1, \dots, x_n \in \mathbb{F}$ , the **Vandermonde matrix** is

$$V(x_1, \dots, x_n) := \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{pmatrix}$$

**Determinant.**

$$\det V(x_1, \dots, x_n) = \prod_{1 \leq i < j \leq n} (x_j - x_i)$$

**Invertibility.**

$$V(x_1, \dots, x_n) \text{ is invertible}$$

$$\iff x_1, \dots, x_n \text{ are pairwise distinct}$$

**Consequences.**

- If  $x_i = x_j$  for some  $i \neq j$ , then two rows are equal  $\Rightarrow \det V = 0$
- If all  $x_i$  are distinct, then  $\operatorname{rank}(V) = n$
- Appears naturally in polynomial interpolation

## 15. Permutations

A **permutation** of  $\{1, \dots, n\}$  is a bijection

$$\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}.$$

The set of all permutations is denoted  $S_n$ .

**Inversions.** An inversion of  $\sigma$  is a pair  $(i, j)$  such that

$$i < j \quad \text{and} \quad \sigma(i) > \sigma(j).$$

Let  $\operatorname{inv}(\sigma)$  be the number of inversions of  $\sigma$ , sometimes denoted  $S(\sigma)$ .

**Sign of a permutation.**

$$\operatorname{sgn}(\sigma) := (-1)^{\operatorname{inv}(\sigma)}.$$

- $\sigma$  is *even* if  $\operatorname{sgn}(\sigma) = 1$
- $\sigma$  is *odd* if  $\operatorname{sgn}(\sigma) = -1$

**Properties.**

- $\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}(\sigma)\operatorname{sgn}(\tau)$
- A transposition changes the sign
- $|S_n| = n!$

**Determinant (Leibniz formula).** For

$$A = (a_{ij}) \in \mathbb{F}^{n \times n},$$

$$\det(A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^n a_{i, \sigma(i)}.$$

**Consequences.**

- If two rows (or columns) are equal, every term cancels  $\Rightarrow \det(A) = 0$
- Swapping two rows multiplies  $\det(A)$  by  $-1$
- Linearity of determinant follows from the formula

## 16. Block Matrices

**Definition.** A *block matrix* is a matrix partitioned into submatrices:

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix},$$

where the blocks have compatible sizes.

**Block operations.** If dimensions agree, block matrices behave like ordinary matrices:

- Addition and scalar multiplication are blockwise
- Matrix multiplication follows block multiplication rules

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} E & F \\ G & H \end{pmatrix} = \begin{pmatrix} AE + BG & AF + BH \\ CE + DG & CF + DH \end{pmatrix}$$

**Block diagonal matrices.**

If

$$A = \begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix},$$

$$\det(A) = \det(A_1) \det(A_2),$$

$$A \text{ invertible} \iff A_1, A_2 \text{ invertible.}$$

**Block triangular matrices.** If

$$A = \begin{pmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{pmatrix},$$

then

$$\det(A) = \det(A_{11}) \det(A_{22}),$$

$$\operatorname{rank}(A) = \operatorname{rank}(A_{11}) + \operatorname{rank}(A_{22}).$$

**Schur complement.** Let

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}, \quad A_{11} \text{ invertible.}$$

The *Schur complement* of  $A_{11}$  in  $A$  is

$$S := A_{22} - A_{21}A_{11}^{-1}A_{12}.$$

- $A$  invertible  $\iff A_{11}$  and  $S$  invertible
- $\det(A) = \det(A_{11}) \det(S)$

**Block inverse formula.** If  $A_{11}$  and  $S$  are invertible, then

$$A^{-1} = \begin{pmatrix} A_{11}^{-1} + A_{11}^{-1}A_{12}S^{-1}A_{21}A_{11}^{-1} & -A_{11}^{-1}A_{12}S^{-1} \\ -S^{-1}A_{21}A_{11}^{-1} & S^{-1} \end{pmatrix}.$$

**Solving block systems.** For

$$\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} b \\ c \end{pmatrix},$$

solve:

1.  $A_{11}x = b - A_{12}y$
2.  $(A_{22} - A_{21}A_{11}^{-1}A_{12})y = c - A_{21}A_{11}^{-1}b$

**Implementation tips.**

- Use block structure to reduce computation
- Common in Gaussian elimination, LU decomposition, and numerical methods
- Essential for large sparse systems and matrix factorizations